

UMP contd.

Monotone likelihood ratio-- Examples

Suppose $X_i \sim \text{ber}(p)$ $i = 1, 2, \dots, n$. Consider $T = \sum_{i=1}^n x_i$.
Here $f_n(\mathbf{t} \mid p)$ has MLR property.

Suppose $X_i \sim \mathcal{N}(\mu, \sigma^2)$. Here μ is unknown but σ^2 is known. Consider $T = \frac{1}{n} \sum_{i=1}^n x_i$.
Here $f_n(\mathbf{t} \mid \mu)$ has MLR property.

Suppose $X_i \sim \text{exp}(\theta)$ $i = 1, 2, \dots, n$. Consider $T = \sum_{i=1}^n x_i$.
Is $f_n(\mathbf{t} \mid \theta)$ has MLR property ?

Monotone likelihood ratio-- Examples

Suppose $X_i \sim \text{ber}(p)$ $i = 1, 2, \dots, n$. Consider $T = \sum_{i=1}^n x_i$.

Here $f_n(\mathbf{t} | p)$ has MLR property.

$$\frac{f_n(\mathbf{x} | p_2)}{f_n(\mathbf{x} | p_1)} = \left[\frac{p_2(1-p_1)}{p_1(1-p_2)} \right]^y \left(\frac{1-p_2}{1-p_1} \right)^n.$$

Suppose $X_i \sim \mathcal{N}(\mu, \sigma^2)$. Here μ is unknown but σ^2 is known. Consider $T = \frac{1}{n} \sum_{i=1}^n x_i$.

Here $f_n(\mathbf{t} | \mu)$ has MLR property.

$$f_n(\mathbf{x} | \mu) = \frac{1}{(2\pi)^{n/2} \sigma^n} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right].$$

Therefore, for every two values μ_1 and μ_2 such that $\mu_1 < \mu_2$,

$$\frac{f_n(\mathbf{x} | \mu_2)}{f_n(\mathbf{x} | \mu_1)} = \exp \left\{ \frac{n(\mu_2 - \mu_1)}{\sigma^2} \left[\bar{x}_n - \frac{1}{2}(\mu_2 + \mu_1) \right] \right\}.$$

Monotone likelihood ratio-- Examples

Suppose $X_i \sim \exp(\theta)$ $i = 1, 2, \dots, n$. Consider $T = \sum_{i=1}^n x_i$.

Is $f_n(\mathbf{x} | \theta)$ has MLR property?

$$f_n(\mathbf{x}|\theta) = \begin{cases} \theta^n \exp(-\theta \sum_{i=1}^n x_i) & \text{for all } x_i > 0, \\ 0 & \text{otherwise.} \end{cases}$$

For $0 < \theta_1 < \theta_2$, we have

$$\frac{f_n(\mathbf{x}|\theta_2)}{f_n(\mathbf{x}|\theta_1)} = \left(\frac{\theta_2}{\theta_1}\right)^n \exp\left([\theta_1 - \theta_2] \sum_{i=1}^n x_i\right),$$

UMP

Let $\mathcal{C}$ be a class of test for testing $H_0 : \theta \in \Theta_0$ vs $H_a : \theta \in \Theta_0^c$. A test in class $\mathcal{C}$, with power function $\pi(\theta)$, is *uniformly most powerful* (UMP) class $\mathcal{C}$ test if $\pi(\theta) \geq \pi'(\theta)$ for every $\theta \in \Theta_0^c$ and every $\pi'(\theta)$ that is a power function of some test in class $\mathcal{C}$.

$$\mathcal{C} = \{ \text{all } \phi(x) \text{ of level } \alpha \}$$

Theorem (Karlin - Rubin) : Suppose:

$$X \text{ has MLR in } T \text{ and } \phi(x) = \begin{cases} 1 & T(x) < c \\ \gamma & T(x) = c \\ 0 & T(x) > c \end{cases}.$$

Also suppose $\alpha = E_{\theta_0} \phi(x)$. Then the test function $\phi(x)$ is a UMP for $H_0 : \theta \leq \theta_0$ $H_1 : \theta > \theta_0$

What it says: If there is a test procedure which rejects null hypothesis H_0 whenever $T(x) \leq c$ for the hypothesis $H_0 : \theta \leq \theta_0$ $H_1 : \theta > \theta_0$ then it is a UMP with level of significance $E_{\theta_0} \phi(x)$

UMP Example